

Dataset:

Order_ID	Customer_ID	Sales_Amount	Order_Date
O101	C001	4500	12-01-2024
O102	C002	Null	15-01-2024
O103	C003	3200	2024/01/18
O101	C001	4500	12-01-2024
O104	C004	Three Thousand	20-01-2024
O105	C005	5100	25-01-2024

1. Data Understanding

Identify all data quality issues present in the dataset that can cause problems during data loading.

Answer:

The given data set contains several data quality issues that can create problems during the ETL process.

1.Duplicate Primary Key values:The Order_ID columns contains duplicate values.The value **O101** appears two times in the dataset.

2.Missing values:The Sales_Amount column contains missing value **NULL** for record **O102**.Missing values create problems in calculations ,reporting and business analysis.

3.Invalid Data Type:The Sales_Amount column should contain numeric values only,but record O104 contains the text value **"Three thousand"** instead of number.

4.Inconsistent Date Formats:The Order_Date column contains multiple date formats:

- 12-01-2024
- 15-01-2024
- 2024/01/18

Duplicate records and inconsistent formatting reduce data reliability and may generate incorrectbusiness reports.

Therefore, the data is not clean and requires preprocessing before loading into the database.

2. Primary Key Validation

Assume Order_ID is the Primary Key.

- a) Is the dataset violating the Primary Key rule.
- b) Which record(s) cause this violation?

Answer:

a). Yes, the dataset violates the Primary Key rule because a Primary Key must contain unique values without duplication.

In this dataset, the value **O101** appears more than once.

b). The following records cause the violation:

The Order_ID **O101** has two records with same Order_Id, making the Primary Key non-unique.

If the data is loaded into a database table where Order_ID is defined as Primary Key, the database will reject the duplicate record and the loading process may fail.

3. Missing Value Analysis

Which column(s) contain missing values?

- a) List the affected records
- b) Explain why loading these records without handling missing values is risky.

Answer:

a). The Dataset contains missing values in the Sales_Amount column.

Sales_Amount Column is missing values. It is denoted by **NULL**.

b). Loading records with the missing values is risky because:

1. **Incorrect Calculations:** Total sales, average sales and revenue reports may become inaccurate.

2. **Reporting Errors:** BI dashboards may show incomplete or misleading KPIs.

3. **Data Integrity Problems:** Important business information becomes incomplete.

4.ETL Failures:If the database column does not allow NULL values,the loading process may fail.

5.Poor Decision Making:Management decisions based on incomplete sales data may become incorrect.

Therefore,missing values should be handled before loading the dataset into the database.

4.. Data Type Validation

Identify records where Sales_Amount violates expected data type rules.

- a) Which record(s) will fail numeric validation?
- b) What would happen if this dataset is loaded into a SQL table with Sales_Amount as DECIMAL?

Answer:

a).The following record will fail numeric validation:

With Order_ID - O104 And Sales_Amount - Three Thousand.

The value "Three Thousand" is stored as text instead of a numeric value.

b).If the Sales_Amount column is defined as decimal in SQL.

- The database will attempt to convert the value into a number.
- The text "Three Thousand" Cannot be converted into Decimal format.
- The insertion process may fail with a data type conversion error.
- Some ETL tools may reject the entire row or stop the loading process completely.

Therefore,invalid datatypes must be corrected before loading the data.

5. Date Format Consistency

The Order_Date column has multiple formats.

- a) List all date formats present in the dataset
- b) Why is this a problem during data loading?

Answer:

a).The dataset contains the following date formats:

1.DD-MM-YYYY

- Exaple:12-01-2024.

2.YYYY/MM/DD

- Example:2024/01/18

b).Different date formats create several problems during ETL Processing.

1.**Incorrect Date Interpretation:**The system may misunderstand the actual date format.

2.**Loading Errors:**Databases expecting a single date format may reject incosistent records.

3.**Sorting Issues:**Reports based on dates may produce incorrect order.

4.**Filtering Problems:**Queries using date ranges may return inaccurate results.

5.**BI Reporting Errors:**Dashbaords may display incorrect trends and monthly summeries.

6.Load Readiness Decision

Based on the dataset condition:

a) Should this dataset be loaded directly into the database? (Yes/No)

b) Justify your answer with at least three reasons

Answer:

a)No,the dataset should not be loaded directly into the databse.

b).The dataset should not be loaded directly because of the following:

1.**Duplicate PrimaryKey:**The Order_ID value O101 is duplicated,which violates databse integrity rules.

2.**Missing Values:** The Sales_Amount value for record O102 is NULL,leading to incomplete sales information.

3.**Invalid Data Formats:**The value "Three Thousand" in Sales_Amount is non-numeric and will failvalidation.

4.**Inconsistent Date Formats:**Different date formats may create loading and reporting problems.

5.**Risk of Incorrect BI Reports:**Unclean data can generate inaccurate dashboards and KPIs.

Hence,data cleaning and validation are necessary before loading the dataset.

7.Pre-Load Validation Checklist

List the exact pre-load validation checks you would perform on this dataset before loading.

Answer:

- Check for duplicate Priimary Key vaues in Order_ID.
- Verify that all mandatory fields contain values.
- Validate that Sales_Amount contains only numeric values.
- Check for NULL or missing values.
- Standardize all date formats into a single format.
- Verify column data types with databse schema.
- Remove Duplicates records.
- Validate date correctness and invalid dates.
- Ensure all records follow business rules and constraints.
- Perform sample testing before final loading.

Thee checks help ensure accurate and reliable data loading.

8.. Cleaning Strategy

Describe the step-by-step cleaning actions required to make this dataset load-ready.

Ansswer:

The following step-by-step cleaning process should be performed to make the dataset load:

1.Identify Duplicate Records: Find duplicate Order_ID values such as **O101**.

2.Remove or Correct Duplicates:Delete unnecesssary duplicate records or assign unique IDs.

3.Handle Missing Values: Replace NULL values iin Sales_Amount using;

- Average Value,
- Default value,
- Business rules.

4.Correct Invalid Data Tyeps: Convert "Three Thousand" Into numeric format such as 3000.

5.Standardize Date Formats:Convert all dates into a single format such as YYYY-MM-DD.

6. Validate Data Types: Ensure numeric columns contain numbers and date columns contain valid dates.

7. Perform Quality Checks: Recheck for duplicates, NULL values and invalid entries.

8. Load Clean Data: After successful validation load the cleaned dataset into the target database.

This cleaning process improves data accuracy and prevents loading failures.

9. Loading Strategy Selection

Assume this dataset represents daily sales data.

a) Should a Full Load or Incremental Load be used?

b) Justify your choice.

Answer:

a). Incremental Load should be used.

b). Justify your choice.

Incremental loading is more suitable because:

1. Faster Processing: Only newly added daily records are loaded instead of reloading the entire dataset.

2. Reduced System Load: Database resources and ETL processing time are minimized.

3. Efficient For Daily Sales Data: Daily sales data continuously grows, making incremental loading more practical.

4. Lower Risk of Duplicate Data: Only changed or new records are processed.

5. Better Performance: BI dashboards and reporting systems can update quickly with new data.

Therefore, Incremental Load is the best strategy for daily sales datasets.

10. BI Impact Scenario

Assume this dataset was loaded without cleaning and connected to a BI dashboard.

a) What incorrect results might appear in Total Sales KPI?

b) Which records specifically would cause misleading insights?

c) Why would BI tools not detect these issues automatically?

Answer:

a).The Total Sales KPI may show incorrect results because:

- Duplicate records may increase total sales.
- Missing values may reduce total revenue.
- Invalid text values may be ignored during calculations,
- Incorrect dates may place sales in wrong periods.

As a result,the dashboards will display inaccurate business performance.

b).The following records would create misleading insights:

Description	ID
Duplicate record	0101
Missing Sales Amount	0102
Invalid Numeric value	0104
Inconsistent Date format	0103

c).BI tools mainly visualize the data provided to them.They do not always prform advanced data validation automatically because:

- BI tools assume source data is already cleaned.
- Duplicate records may still appear valid.
- Text values may be ignored silently during calculations.
- NULL values may not generate visible errors.
- Incorrect dates may still load without warning.

Therefore,ETL data cleaning is very important before connectiong data to BI dashboards.